

SRM Institute of Science and Technology

Course Code: 21CSC302J | Course Name: COMPUTER NETWORKS

Assignment 2

Unit-2: Addressing

Instructions: Answer all questions. Draw neat diagrams wherever necessary. Show complete working for numerical questions.

- Q1.** Explain IPv4 addressing and the concept of address space with its structure (network ID and host ID).
- Q2.** Explain Classful IP addressing. List the address ranges of Class A, B, C, D and E, with the default subnet mask for each.
- Q3.** What is a subnet mask? Explain, with an example, how it is used to identify the network and host portion of an IP address.
- Q4.** Explain Classless Addressing (CIDR) and how it overcomes the limitations of classful addressing. Give one CIDR example.
- Q5.** Explain Variable Length Subnet Masking (VLSM). State its advantage over Fixed Length Subnet Masking (FLSM) with a suitable example.
- Q6.** Explain the concept of supernetting with a numerical example of combining two or more Class C networks.
- Q7.** What is Network Address Translation (NAT)? Explain its types - Static NAT, Dynamic NAT and PAT (Port Address Translation).
- Q8.** Given the network address 192.168.10.0/24, design subnets for four departments requiring 50, 30, 20 and 10 hosts respectively using VLSM. Show all working.
- Q9.** Differentiate between Static Routing and Default Routing. Write the generic configuration steps for each on a router.
- Q10.** Explain the steps involved in configuring NAT on a router. Illustrate with example configuration commands.

--- End of Assignment ---